Few-Shot Classification with Semantic Augmented Activators

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Abstract. Metric-based methods predict class labels by measuring the distance between a few given samples, often failing to preserve more useful semantic details in their vectorial representations. In this paper, we propose Semantic Augmented Activators (SAA), which are generated based on the variance of the intra-set samples in an unsupervised manner, to enhance the discriminability of feature vectors with more class-related semantic information. This generation process does not rely on any learnable parameters. Meanwhile, to align the SAA preferred to operate in the intra-set and sufficiently leverage the finite samples, we treat the Self-Cross loss as an auxiliary loss, which bi-directionally complements the limitations of the traditional loss function. Additionally, we introduce Map-To-Cluster, a transductive module to map the SAA-enhanced features to a lower-dimensional embedding space. This encourages proximity among similar samples and separation among dissimilar samples. The resulting methods are lightweight and computationally efficient. Our methods demonstrate competitive performance on the *mini*-ImageNet and *tiered*-ImageNet benchmarks, and achieve outstanding results in Cross-Domain Few-Shot classification.

Keywords: Few-shot learning · Metric-based learning · Transductive inference

1 Introduction

Traditional convolutional neural classifiers rely on abundant annotated image sets, such as ImageNet [18], and COCO [12], to achieve brilliant performance. However, acquiring such extensive human-labeled datasets is extremely challenging in reality. Inspired by the remarkable ability of humans to classify objects

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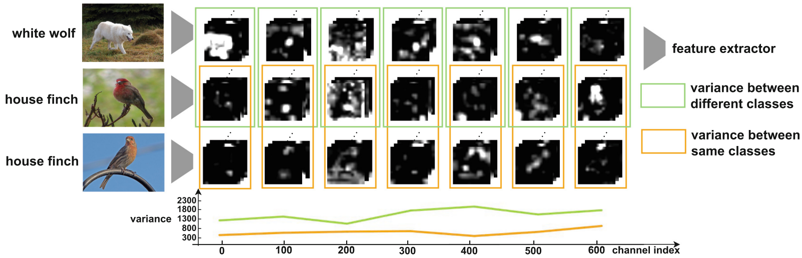


Fig. 1. Variance of channels: We calculate the mean variances for the same class and different classes per 100 channels. The variances of channels from the same class are obviously lower than those from different classes.

with a small amount of data and quickly adapt to new tasks, few-shot learning emerged and grew in popularity. Few-shot learning aims to predict image class labels with only a few annotated samples. Nevertheless, the limited support images fail to provide sufficient information for the model to learn the underlying semantic representation, and the model needs to recognize numerous novel classes that were not present in the training set.

Metric-based methods, as one of the solution branches, utilize distance functions to measure the similarity between vector representations of images in the embedding space for predicting class labels. Commonly used distance functions include cosine similarity [24], Euclidean distance [21], or learnable modules [22]. In general, performance heavily relies on the class-related semantic content encoded in the representations. Therefore, enhancing the semantic information within single vectorial representations becomes a crucial challenge for researchers. Previous approaches such as feature reconstruction [9,27] focus on capturing similar semantic information between support and query samples, but they often require additional attention modules or parameters. Alternatively, incorporating prior knowledge [7] as supplementary semantic information to feature vectors is also explored, but it asks a weight generator for help. In this paper, we aim to analyze the distribution within features without introducing additional modules, activating relevant semantic information to obtain discriminative feature vectors.

SPP-net [8] has demonstrated that filters of deep convolutions can be selectively activated by specific semantic content, and PWA [28] utilizes filters of convolutional layers as part detectors to highlight discriminative regions of samples while suppressing background noise. These approaches show that the layers of feature maps contain valuable cues for understanding objects. As shown in Fig. 1, the variances between two samples from the same class are significantly lower than that from different classes in the same channels, illustrating that the variance of channels can capture the discrepancy in semantic information and exhibit varying degrees of response to class-related semantic information. Motivated by the above experiment, we utilize the channels of feature maps selected based on their variances as Semantic Augmented Activators (SAA). These activators allow us to increase the proportion of distinguishable regions, thereby augmenting the efficacious semantic information and the representational capacity of the features. It is noted that this is an unsupervised process.

SAA focuses on capturing intra-set features. However, the traditional loss function only unidirectionally enforces the model to learn inter-set features and limits the utilization of the given samples. To address this limitation, we propose Self-Cross loss (SC loss) as an auxiliary loss. If the query features can obtain sufficient semantic information from support prototypes, it can also provide classification prediction for support features in the reverse direction. The support prototypes not only interact with query features but also with support features. Similarly, the query prototypes also interact with both query features and support features. This bi-directional loss strengthens the connections between samples and promotes few-shot classification. Furthermore, taking inspiration from transductive methods such as Poodle [10] and EASE [31], which leverage the entire query set, we introduce the transductive module called Map-To-Cluster (MTC) based on cluster theory. MTC reduces the dimension of query features, aiming to filter outlying contents and focus on the essential semantic information. MTC complements existing works in few-shot classification, further improving the performance.

In summary, this paper makes the following contributions:

1. We propose SAA, generated by an unsupervised mechanism, incorporating useful semantic information into the discriminative vectorial representations, without any learnable parameters.
2. We introduce SC loss to extract features not only of the inter-set but also of the intra-set and constraint model to hoist generalization with bi-directional formulas.
3. We utilize the transductive module MTC, which closes the distance between congenetic samples while increasing the separation between samples from different classes.

2 Related Work

2.1 Few-Shot Classification

Few-shot classification aims to predict novel class labels using a limited number of annotated samples. Generally, they are divided into two main branches: optimization-based methods and metric-based methods. Optimization-based approaches [6,16] learn better initial parameters to facilitate subsequent training. On the other hand, metric-based methods [7,9,21,22,24,27] emphasize measuring the distance between samples. Several excellent works make splashes in the metric-based branch. For example, ProtoNet [21] proposes the concept of prototypes and utilizes more representative prototypes for comparing Euclidean distance with the samples, while Matching network [24] introduces LSTM and uses cosine similarity to make predictions. In addition to traditional distance functions, a learnable model [22] can also be introduced.

Furthermore, CTX [5] achieves more generalized feature representations through SimCLR [1]. CAN [9] and FRN [27] reconstruct query samples so that more similar samples have more parallel features between the support set and query set, with the former focusing on attention maps and the latter employing ridge regression. Classification-weight generator [7] effectively incorporates past knowledge through the attention mechanism while learning novel vector representations. In contrast to the above methods that emphasize exploring the relationship between the features of the support set and the query set, our methods focus on analyzing the inherent properties of the intra-set samples, capturing class-related semantic information without the need for complex computations.

2.2 Transductive Inference

Transductive inference involves treating query samples, which are typically considered unlabeled during training, as a whole and making predictions for them. It can boost the performance of few-shot classification. CAN [9] obtains more representative features by augmenting the support samples with the query samples. Poodle [10] and ConFT [4] wield the out-of-distribution samples to improve few-shot generalization. In contrast, our plug-and-play MTC does not introduce extra samples but instead clusters the existing data. This process focuses on the query samples themselves, without relying on support samples and corresponding labels for assistance.

3 Method

In this section, we describe the mechanism of SAA and explain the details of SC loss. Followed by presenting the transductive module MTC used to filtrate SAA-enhanced features. An overview is provided in Fig. 2.

3.1 Preliminaries

We first define some notations used in the few-shot classification scenario. Designating (x_i, y_i) as an image and its corresponding label. The base training set utilized for the pre-training stage is denoted as $D_b = \{(x_i, y_i)\}_{i=1}^{N_b}$. Following we expect to predict labels of unseen samples in the query set $D_q = \{(x_i^q, y_i^q)\}_{i=1}^{N_q}$ using the support set $D_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$. Here, the variables N_b, N_s, and N_q represent the respective number of sample pairs in each set. In a few-shot classification task episode involving N classes with K samples per class, we refer to this as N-way K-shot. Note that the labels of the query set are only used for verification, and therefore we treat the query samples as unlabeled.

While considering scenarios where both training and validation are in the same domain, we also assess the generalization of our methods in a cross-domain scenario. In this scenario, we train the model in one domain and then test its performance in another domain.

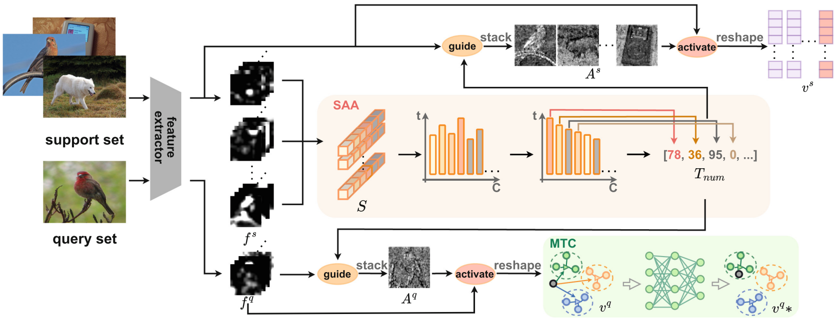


Fig. 2. Flow chart of our methods. First, we calculate the variances of channels and sort them in descending order. This allows us to determine the sequence numbers of channels, which are used to select the top L channels for each sample and stack them to generate the activators. Then, MTC reduces the dimension of SAA-enhanced feature vectors to output the final prediction.

3.2 Semantic Augmented Activators

We propose SAA with the expectation of enhancing feature vectors to stockpile more class-related semantic information for accurate prediction of query labels y^q. Given support samples x^s, we feed them into the embedding module F_θ to obtain feature maps $f^s \in \mathbb{R}^{N_s \times C \times W \times H}$.

First, we perform spatial aggregation of the feature maps for each support sample, resulting in the computation of the summing descriptors $S \in \mathbb{R}^{N_s \times C}$ for each channel. This aggregation is achieved by sum-pooling the feature maps f^s.

$$S = \sum_{w=0}^{W} \sum_{h=0}^{H} f^s(w,h) \tag{1}$$

Variance can measure the degree of differentiation between samples within the same channel, which also indirectly indicates the response of channels to specific categories. A higher variance implies that the corresponding channel contains more discriminative semantic information.

$$T = \frac{1}{N_s} \sum_{i=1}^{N_s} (S_i - \frac{1}{N_s} \sum_{r=1}^{N_s} S_r)^2 \tag{2}$$

According to the calculated variances $T = \{t_1, t_2, ..., t_C\}$, we sort them in descending order to obtain the corresponding channel sequence numbers T_{num}. We then select L channels for each support sample based on T_{num} to stack L-activators, denoted as $A^s \in \mathbb{R}^{N_s \times W \times H}$. These activators highlight the specific class semantic regions while suppressing the background noise, leading to a strong response to the objects.

$$A_r^s = \sum_{l}^{L} f_r^s[T_{num}[l]], r = 1, 2, ..., N_s \tag{3}$$

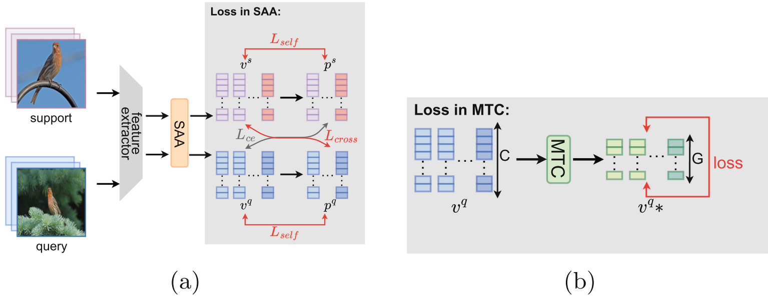


Fig. 3. The visual representation of the loss function. (a) describes the SC loss. The SC loss is specifically designed for inductive inference and assists the feature extractor in adequately exploiting the given samples. (b) illustrates the loss function used to optimize the MTC, which leverages transductive inference to boost performance.

It is worth noting that the query activators A^q are selected from the query feature maps based on the same channel sequence numbers T_{num} obtained from the support set. This can enhance the interaction of the same semantic information and the consistency of the same class information between different sets. The support weights $W^s \in \mathbb{R}^{N_s \times W \times H}$ is the normalized coefficient consulted by A^s, while the query weights $W^q \in \mathbb{R}^{N_q \times W \times H}$ incorporates the semantic information learned from the support set, facilitating inter-set knowledge sharing.

$$\begin{aligned} W^s &= \mathbb{M}(\frac{A^s}{\sum_w^W \sum_h^H A^s(w,h)}) \\ W^q &= \mathbb{M}(\alpha \times \mathbb{G}(W^s) + \beta \times \frac{A^q}{\sum_w^W \sum_h^H A^q(w,h)}) \end{aligned} \tag{4}$$

where $\mathbb{G}(\cdot)$ is the repeat operator, constraining that the dimensions of the operands are the same, $\mathbb{M}$ denotes a spatial-wise sigmoid, α and β are parameters of power-scaling. Specifically, we set $\alpha = 0.5$ and $\beta = 1.0$ for the *mini*-ImageNet, while $\alpha = 0.1$ and $\beta = 1.0$ for the *tiered*-ImageNet.

Following we weigh the initial representations f^s and f^q with W^s and W^q to highlight the discriminative regions of images and enrich efficacious semantic information for the subsequent feature vectors. We reshape the activated feature maps with global average-pooling to obtain the final augmented feature vectors $v^s \in \mathbb{R}^{N_s \times C}$ and $v^q \in \mathbb{R}^{N_q \times C}$. For getting the corresponding support prototypes $p^s \in \mathbb{R}^{N \times C}$, we calculate the average of v^s belonging to the same class.

$$\begin{aligned} v^s &= \mathbb{P}(W^s f^s) \\ p_n^s &= \frac{1}{|K|} \sum_{k=1}^{K} v_{n,k}^s, n = 1, 2, ..., N \end{aligned} \tag{5}$$

where $\mathbb{P}(\cdot)$ is the global average-pooling.

3.3 Self-Cross Loss

For considering the features of the intra-set involved in SAA and to maximize the utilization of the given samples, we treat SC loss as an auxiliary loss to bi-directionally constrain the model. As shown in Fig. 3(a), in addition to the traditional loss L_{ce}, we introduce a new loss called SC loss L_{sc}, which is composed of two components: L_{cross} and L_{self}.

If the query feature vectors v^q learned from SAA are representative, the query prototypes p^q generated from them contain sufficient categorical discriminative semantic information, predicting support feature vectors v^s in reverse and deepening the association of samples from the same class.

$$L_{cross} = CE(d(v^s, p^q), y^s) \tag{6}$$

where $CE(\cdot)$ is the cross-entropy loss, $d(\cdot)$ is the Euclidean distance and p^q are the query prototypes. Meanwhile, to encourage alignment between samples from the same class within the same set, we calculate the loss L_{self} by comparing the feature vectors to their corresponding prototypes in each set.

$$L_{self} = CE(d(v^s, p^s), y^s) + CE(d(v^q, p^q), y^q) \tag{7}$$

Traditional loss and SC loss compose the final loss to further improve the performance of the framework.

$$L = L_{ce} + \underbrace{\gamma_1 L_{self} + \gamma_2 L_{cross}}_{L_{sc}} \tag{8}$$

where L_{ce} is the original loss function. $\gamma_1 = 10.0$ and $\gamma_2 = 1.0$ are parameters of power-scaling for SC loss.

3.4 Map-To-Cluster

We introduce the MTC to further bring similar samples closer together and separate dissimilar samples to boost performance. For MTC, we do not rely on the class labels but only leverage the knowledge learned from SAA.

We map the SAA-enhanced query features through MTC to the G-dimensional ($G < C$) embedding space, denoted as v^q*. MTC is composed of two linear layers with the LeakyReLU function. According to the nearest neighbor [26], the means of query features are taken as the initial cluster centers. The congenetic samples move closer to the centers of the same class in the learned G-dimension embedding space. The final prediction depends on the distances between the samples and cluster centers, and the MTC uses a separate loss function shown in Fig. 3(b) for continuous optimization.

$$P(n|x_i^q) = \frac{e^{\| v_{n,i}^q * - \frac{1}{|N_q|} \sum_{z=1}^{N_s} v_{n,z}^q * \|^2}}{\sum_{n'=1}^{N} e^{\| v_{n,i}^q * - \frac{1}{|N_q|} \sum_{z=1}^{N_s} v_{n',z}^q * \|^2}} \tag{9}$$

Table 1. Performance of selected competitive few-shot classification methods on *mini*-ImageNet and *tiered*-ImageNet. ∘ means transductive inference. ♡ means knowledge distillation.

Model	Backbone	*mini*-ImageNet		*tiered*-ImageNet	
		1-shot	5-shot	1-shot	5-shot
ProtoNet [21,29]	ResNet-12	62.39 ± 0.21	80.53 ± 0.14	68.23 ± 0.23	84.03 ± 0.16
MatchNet [24,30]	ResNet-12	63.08 ± 0.80	75.99 ± 0.60	68.50 ± 0.92	80.60 ± 0.71
MetaOptNet [11]	ResNet-12	62.64 ± 0.61	78.63 ± 0.46	65.99 ± 0.72	81.56 ± 0.53
CAN [9]	ResNet-12	63.85 ± 0.48	79.44 ± 0.34	69.89 ± 0.51	84.23 ± 0.37
MetaBaseline [3]	ResNet-12	63.17 ± 0.23	79.26 ± 0.17	68.62 ± 0.27	83.29 ± 0.18
DSN [20]	ResNet-12	62.64 ± 0.66	78.83 ± 0.45	66.22 ± 0.75	82.79 ± 0.48
Neg-Cosine [13]	ResNet-12	63.85 ± 0.81	81.57 ± 0.56	–	–
Deepemd [30]	ResNet-12	65.91 ± 0.82	82.41 ± 0.56	71.16 ± 0.87	86.03 ± 0.58
FRN [27]	ResNet-12	66.45 ± 0.19	82.83 ± 0.13	71.16 ± 0.22	86.01 ± 0.15
P-Transfer [19]	ResNet-12	64.21 ± 0.77	80.38 ± 0.59	–	–
CAN+T∘ [9]	ResNet-12	67.19 ± 0.55	80.64 ± 0.35	73.21 ± 0.58	84.93 ± 0.38
LaplacianShot∘ [32]	ResNet-18	72.11 ± 0.19	82.31 ± 0.14	78.98 ± 0.21	86.39 ± 0.16
poodle∘ [10]	ResNet-12	74.21	83.71	78.72	86.57
poodle∘ ♡ [10]	ResNet-12	77.56	**85.81**	79.97	86.96
SAA(ours)	ResNet-12	65.39 ± 0.61	81.28 ± 0.43	70.03 ± 0.72	84.53 ± 0.49
+MTC(ours)∘	ResNet-12	76.82 ± 0.41	82.93 ± 0.38	84.41 ± 0.40	86.15 ± 0.36
+MTC(ours)∘ ♡	ResNet-12	**82.79 ± 0.38**	**85.64 ± 0.31**	**86.47 ± 0.37**	**87.28 ± 0.35**

4 Experiments

In this section, we give the implementation details of our methods and conduct extensive experiments to demonstrate the performance of our methods under different scenarios. Finally, we supplement ablation experiments to illustrate the effectiveness of each part of our methods.

4.1 Set up

On the general few-shot classification, we evaluate our methods on *mini*-ImageNet [24] and *tiered*-ImageNet [17]. Instead of training directly from scratch using episodes, we pre-training like [29] on the training set. To verify the generalization of our method, we also test the effectiveness of our method on cross-domain problems in which the model trained on *mini*-ImageNet is evaluated on CUB [25].

***mini*-ImageNet** chooses 60000 images from ImageNet [18] and 600 images per class, with 100 classes including 64 training, 16 validation, and 20 test classes.

***tiered*-ImageNet** is the subset of the ImageNet, consisting of 351 training, 97 validation, and 160 test classes with 779165 images.

CUB is the fine-grained dataset with 11788 images. We randomly split the 200 classes into 100 training, 50 validation, and 50 test.

Implementation Details: We treat ResNet-12 as the feature extractor and ProtoNet as the baseline. During pre-training, we adopt cross-entropy loss to train our model with a batch size of 128. For meta-training, we use SGD optimizer with a fixed learning rate of 3e-4, while in transductive inference, we utilize Adam optimizer with a fixed learning rate of 1e-4. The number of iterations of MTC optimization is consistent with that of the whole model, without repeating iterations. The final performance we evaluate in 5-way 1-shot and 5-shot settings on 600 random sample episodes with 15 query images of size 84×84.

Table 2. Performance comparison in the cross-domain setting: *mini*-ImageNet → CUB.

Model	Backbone	5-way 1-shot	5-way 5-shot
ProtoNet [2,21]	ResNet-18	–	62.02 ± 0.70
MetaOptNet [11,15]	ResNet-12	44.79 ± 0.75	64.98 ± 0.68
MatchNet+FT [23]	ResNet-10	36.61 ± 0.53	55.23 ± 0.83
RelationNet+FT [23]	ResNet-10	44.07 ± 0.77	59.46 ± 0.71
GNN+FT [23]	ResNet-10	47.47 ± 0.75	66.98 ± 0.68
ConFT [4]	ResNet-10	45.57 ± 0.76	70.53 ± 0.75
SAA(ours)	ResNet-12	48.49 ± 0.60	66.07 ± 0.55
+MTC(ours)	ResNet-12	**63.99 ± 0.47**	**74.20 ± 0.39**

Table 3. The speed comparison per episode (ms) on *mini*-ImageNet.

Methods	meta-training		meta-test	
	5-way 1-shot	5-way 5-shot	5-way 1-shot	5-way 5-shot
ProtoNet	210	250	83	105
Deepemd(qpth)	26,000	>990,000	23,275	>800,000
SAA(ours)	396	451	308	311
+MTC(ours)	424	485	311	325

4.2 Results

General Few-Shot Classification: We show the results of classification with 5-way 1-shot and 5-way 5-shot in Table 1. On *mini*-ImageNet, we find using SAA alone will achieve 3% improvement over amended ProtoNet with 5-way 1-shot, and the performance is even more significant with 14.43% increase when using transductive method MTC to map learned features to G-dimensional embedding space. Although the increase is not very significant on *tiered*-ImageNet, SAA still achieves around 2% improvement, and MTC can even achieve an increase of 16%. 5-way 5-shot shows similar improvement. That means SAA effectively learns discriminative feature vectors with class-related semantic information. And MTC indeed boosts performance by bringing samples of the same classes closer. We also conduct experiments with knowledge distillation, which have outstanding results, matching certain methods using similar additional techniques [10].

Cross-Domain Few-Shot Classification: We evaluate the transferability of our methods on the challenging Cross-domain Few-shot classification. We train the model on *mini*-ImageNet during the pre-train and meta-train stages, and then only test it on the CUB to investigate the discrepancies between the different domains. As shown in Table 2, our methods demonstrate outstanding performance, with MTC being particularly effective in cross-domain variations. We suspect that MTC is trained based on the SAA and not constrained by prior knowledge.

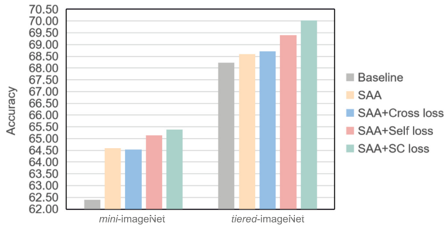
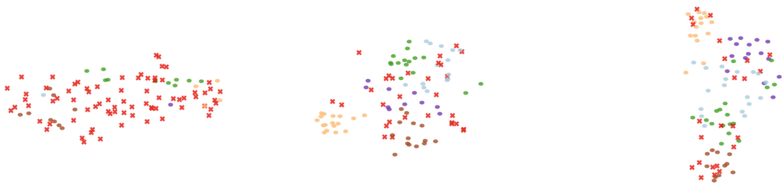


Fig. 4. The accuracy of our method with different SC loss settings on *mini*-ImageNet and *tiered*-ImageNet with 5-way 1-shot.



(a) Inital state (b) After SAA (c) After MTC

Fig. 5. The visualization of MTC on *mini*-ImageNet with 5-way 1-shot. **The red crosses** representing misclassified samples reduce significantly through MTC.

Ablation Experiments: As shown in Table 3, we do speed experiments to verify that our method is lightweight. Even after incorporating MTC, our method takes only slightly more time per episode than the baseline, but at least 23 s faster than Deepemd.

We conduct ablation experiments on the SC loss to further validate its effectiveness using a 5-way 1-shot setting. Figure 4 demonstrates that both the Self loss and Cross loss components of the SC loss contribute to performance improvement. However, it is worth noting that the performance with the Cross loss in the *mini*-ImageNet seems to betray this result. We suspect that the samples in the *mini*-ImageNet belong to the broader categories, making them more separable. In this case, reusing the inter-set finite samples leads to overfitting and potentially harms the performance.

We adopt t-SNE [14] to visualize the dynamic distribution of correct and incorrect samples. As shown in Fig. 5, the incorrect samples represented by **red crosses** are significantly reduced after the introduction of SAA and MTC.

5 Conclusion

We propose Semantic Augmented Activators, learned by an unsupervised mechanism, to highlight the regions of objects while suppressing noise information. This results in feature vectors that contain more discriminative semantic information. To assist the SAA focused more on intra-set, Self-Cross loss as an auxiliary loss to bi-directionally constraints the model. Additionally, we introduce Map-To-Cluster to extract essential semantic information by reducing the dimension of the feature vectors. Our experiments achieve excellent performance on both general few-shot classification and cross-domain problems.

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